

Unit 14, Lesson 3: Calculating Theoretical Probability

Benchmarks

MA.B.DP.2.1 Determine the sample space for a repeated experiment.

MA.B.DP.2.2 Find the theoretical probability of an event related to a repeated experiment.

Learning Target

- ☐ I can calculate the theoretical probability of the outcomes in a repeated experiment.

14.3.1 Warm-Up: Solving Proportions

1. Solve the proportions for the given variable.

a. $\frac{x}{54} = \frac{20}{100}$ $x =$

b. $\frac{x}{6} = 0.5$ $x =$

c. $20\% = \frac{6}{x}$ $x =$

14.3.2 Guided Instruction: Determining Probability

Probability can be used to better understand random processes. One way to analyze a random process is by calculating the theoretical probability of the events in a sample space.



Theoretical probability is a number between 0 and 1 that represents the likelihood of an event in a theoretical model based on a sample space. If all outcomes in the sample space are equally likely, then the theoretical probability of an event is the ratio of the number of outcomes in the event to the number of outcomes in the sample space.

1. **Samaria plays a game that involves rolling two fair six-sided dice.**
 - a. **Determine the outcomes in the sample space for the game.**

- b. Samaria only wins if she rolls a sum of 9 or more. How many of the outcomes will result in a win?
- c. Since each outcome in the sample space is equally likely, a ratio can be created to compare the outcomes with a sum greater than or equal to 9 to the total number of outcomes.

$$P(\text{Sum} \geq 9) = \frac{\text{the number of ways to get a sum} \geq 9}{\text{the total number of outcomes in the sample space}} = \underline{\hspace{2cm}}$$

This ratio represents the probability of Samaria winning.

- d. Write the probability of winning as a fraction, decimal rounded to the nearest hundredth, and percentage rounded to the nearest tenth.

Fraction:

Decimal:

Percentage:

To calculate a theoretical probability, think through the random process and use the sample space to determine the probability of certain events in the sample space.

14.3.3 Guided Practice: Calculating Theoretical Probability

1. Ashley places 3 red marbles, 3 blue marbles, 3 yellow marbles, and 3 orange marbles in a bag.

A representation of the marbles is shown.



- What are the possible outcomes of randomly choosing one marble?
- How many outcomes are in the sample space when choosing two marbles, with replacement?
- Write the sample space for choosing two marbles from the bag, with replacement.
- Are all outcomes equally likely when choosing two marbles, with replacement? Explain your reasoning.
- Let Event A = two marbles of the same color are chosen. How many outcomes are in Event A ?

- f. What is the probability of choosing two marbles of the same color, $P(A)$?

Fraction	Decimal	Percentage

2. A fair coin with one pink side and one blue side and a fair spinner are shown.



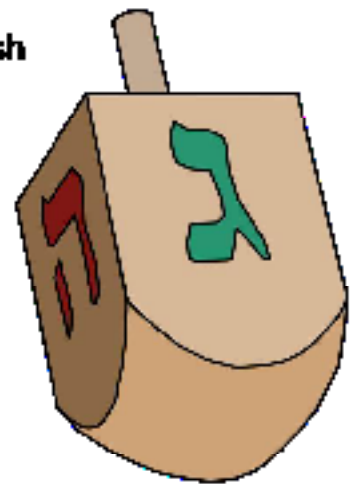
- a. If the coin is tossed and the spinner is spun, how many outcomes are in the sample space?
- b. Write the sample space.
- c. What is the probability that both land on blue?
- d. If the coin is tossed eight times, how many outcomes are in the sample space?
- e. What is the probability that it lands on pink all eight times?

14.3.4 Your Turn!

1. **A dreidel is a four-sided spinning top, played during the Jewish holiday of Hanukkah. Each side has a letter of the Hebrew alphabet: נ (nun), ג (gimel), ה (hei), and ש (shin).**

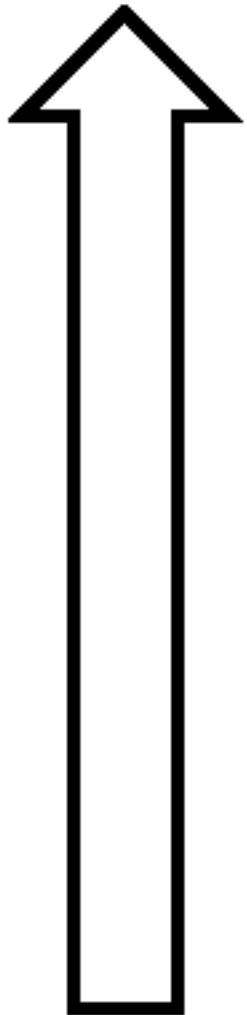
David presents a new game to his friends involving a dreidel and a fair coin. If the dreidel lands on ה (hei) and the coin lands on heads OR the dreidel lands on ג (gimel) and the coin lands on tails, the player wins.

- a. Define the sample space.
- b. How many outcomes result in a win?
- c. What is the probability a player wins?



14.3.5 Wrap-Up: Reflection

1. **Color in the arrow up to the statement that best describes your current understanding of calculating theoretical probability for a two-step experiment.**



I'm so confident, I can explain this to someone else!

I can get to the right answer, but I don't understand well enough to explain it yet.

I understand some of this, but I don't understand all of it yet.

I tried hard and listened, but I am finding this challenging. I will make sure I get help with this in the next lesson.

I do not understand any of this yet. There are things I could do to be a better learner next lesson.